

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	M Poojitha Reddy	Reg. No.:	192472352
Section / Batch:	2024	Date:	27-08-26

Code of Conduct

I, M Poojitha Reddy - 192472352 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M Poojitha Reddy

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. Reference Resolution and Discourse Coherence

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) Identify all the referring expressions and their possible antecedents.

b) Apply the following constraints to resolve the references:

- **Gender and number agreement**
- **Recency**
- **Grammatical role**
- **Semantic compatibility**
- **Discourse coherence**

c) Construct a table showing:

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
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d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

2. Dialog Act Recognition and Response Generation

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

1. Maintain entity coherence using assignment, mistakes, and you.
2. Use a Cause–Effect or Elaboration discourse relation.
3. Include at least two keywords from:
 - review
 - improve
 - confident
 - feedback
4. Response length must be 2–3 sentences.
5. Maintain a positive and polite tone.
6. Do not give contradictory advice.

Tasks:

- a) Identify the dialog acts required.
- b) Identify the important entities that must be maintained.
- c) Generate three possible responses satisfying all constraints.
- d) Compare the responses based on:
 - Coherence
 - Constraint satisfaction
 - Politeness
 - Relevance
- e) Select the best response and justify your answer.
- f) Explain what happens if entity coherence is not maintained.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence:

"Priya sat on the bank and watched the boats moving across the water."

Note:

The word "bank" can refer to:

- A financial institution
- The side of a river

Constraints:

1. Resolve the correct word sense using contextual information.

2. Identify important entities and actions.
3. Represent the meaning using predicate logic.
4. Preserve semantic relationships.
5. Maintain coherence in the generated paraphrase.

Tasks:

- a) Identify the possible meanings of bank.
- b) Select the correct sense and justify your decision using contextual constraints.
- c) Write a predicate logic representation using suitable predicates such as:

sit(x)

bank(y)

beside(x,y)

watch(x,z)

boat(z)

move(z)

water(w)
- d) Generate a simple English paraphrase without ambiguity.
- e) Explain how Word Sense Disambiguation improves Machine Translation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1	4	4	4	3	4	19	58
2	4	4	4	4	4	20	
3	4	3	4	4	4	19	

Faculty In-charge	Course Coordinator	Program Director
T Kumaragurubaran		

Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Reference Resolution and Discourse Coherence

Given Paragraph

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

a) Referring Expressions and Possible Antecedents

Referring Expression	Possible Antecedents
He	Ravi, Arun
him	Ravi, Arun
it	book on Artificial Intelligence
they	Ravi and Arun
the book	book on Artificial Intelligence

b) Applying Constraints

1. Gender and Number Agreement

- **He** → Ravi or Arun (both male, singular)
- **him** → Ravi or Arun (male, singular)
- **it** → book (non-human, singular)
- **they** → Ravi + Arun (plural)

2. Recency

- The nearest previous noun phrase is usually preferred.
- For **He**, Arun is more recent than Ravi.

3. Grammatical Role

- Subjects are preferred antecedents.
- Ravi is subject of the previous sentence.

4. Semantic Compatibility

- A person can search for a book.
- A book cannot help someone.
- "It" matches an object (book).

5. Discourse Coherence

- Since the next sentence says "**Arun helped him find it,**" it is coherent that:
 - Ravi searched for the book.
 - Arun helped Ravi.

c) Reference Resolution Table

Referring Expression	Possible Antecedents	Valid / Invalid	Reason
He	Ravi	Valid	Subject preference and coherent discourse
He	Arun	Possible but weaker	Recency supports Arun, but later sentence suggests Ravi searched
him	Ravi	Valid	Arun helped Ravi
him	Arun	Invalid	Arun cannot help himself in this context
it	Book on AI	Valid	Singular non-human object
it	Library	Invalid	Semantically incorrect
they	Ravi + Arun	Valid	Plural agreement
they	Ravi only	Invalid	Number mismatch
the book	Book on AI	Valid	Same entity

d) Final Coreference Chains

Chain 1: Ravi

Ravi → He → him

Chain 2: Arun

Arun

Chain 3: Book

book on Artificial Intelligence → it → the book

Chain 4: Group

Ravi + Arun → they

e) Paragraph with Pronouns Replaced

"Ravi met Arun at the library. Ravi was looking for a book on Artificial Intelligence. Arun helped Ravi find the book. Later, Ravi and Arun discussed the book before leaving."

f) Ambiguity if "He was looking for a book" Appears Earlier

Example:

"Ravi met Arun at the library. He was looking for a book. Arun helped him find it."

If **He** appears before introducing Arun:

"Ravi met someone at the library. He was looking for a book."

Ambiguity arises because:

- The reader does not know whether **He** refers to Ravi or another person.
- Reference resolution becomes difficult due to missing antecedents.
- Discourse coherence decreases because entities are not introduced before pronouns are used.

This is called an **anaphora resolution problem**.

2. Dialog Act Recognition and Response Generation

a) Required Dialog Acts

1. **Reassure**
 - Reduce the student's anxiety.
 2. **Advise**
 - Suggest constructive actions.
-

b) Important Entities

- **assignment**
- **mistakes**
- **you (student)**
- **review**
- **feedback**

Maintaining these entities ensures discourse coherence.

c) Three Responses

Response 1

"You should not worry too much about your assignment because everyone makes mistakes while learning. You can review your work after receiving feedback, and this will help you improve and feel more confident."

Response 2

"It is completely normal to be concerned about mistakes in your assignment, but submitting it is already a positive step. When you receive feedback, review it carefully because it can help you improve and become more confident in your future work."

Response 3

"Please do not be discouraged by possible mistakes in your assignment, as they are part of the learning process. Reviewing feedback will help you improve your skills, and this progress will make you more confident over time."

d) Comparison

Criteria	Response 1	Response 2	Response 3
Coherence	Good	Excellent	Very Good
Constraint Satisfaction	Yes	Yes	Yes
Politeness	Good	Excellent	Excellent

Criteria	Response 1	Response 2	Response 3
Relevance	Good	Excellent	Very Good

e) Best Response

Best: Response 2

Justification:

- Maintains all entities.
- Uses cause-effect relation:
 - Feedback → Review → Improve → Confidence.
- Polite and encouraging.
- Contains required keywords:
 - review
 - improve
 - feedback
 - confident
- Fully satisfies all constraints.

f) Effect of Losing Entity Coherence

Example:

"Don't worry. Practice more and be happy."

Problems:

- No mention of **assignment** or **mistakes**.
- Response becomes vague.
- User may feel misunderstood.
- Reduced relevance and coherence.

Entity coherence ensures continuity and meaningful interaction.

3. Word Sense Disambiguation (WSD) and Meaning Representation

Sentence:

"Priya sat on the bank and watched the boats moving across the water."

a) Possible Meanings of "Bank"

1. **Financial institution**
 - A place where money is stored.
2. **River bank**

- Land beside a river or lake.

b) Correct Sense and Justification

Correct Sense:

✓ River bank

Contextual Constraints

Context Word	Interpretation
boats	Related to rivers/water
water	Indicates natural environment
sat on	People sit on land, not financial institutions

Therefore, **bank = side of a river.**

c) Predicate Logic Representation

Let:

- **p = Priya**
- **b = river bank**
- **bt = boats**
- **w = water**

Predicates

Priya(p)
Bank(b)
RiverBank(b)
Sit(p)
On(p, b)
Boat(bt)
Water(w)
Move(bt)
Across(b, tw)
Watch(p, bt)

Combined representation:

$Priya(p) \wedge RiverBank(b) \wedge Sit(p)$
 $On(p, b) \wedge Boat(bt)$
 $Water(w) \wedge Move(bt)$
 $Across(b, tw) \wedge Watch(p, bt)$

d) Unambiguous Paraphrase

"Priya sat on the riverbank and watched boats moving across the water."

e) How WSD Improves Machine Translation

Word Sense Disambiguation (WSD) helps translation systems choose the correct meaning of ambiguous words.

Example:

English:

"Priya sat on the bank."

Without WSD:

- "Bank" may be translated as a financial institution.

With WSD:

- Context words such as **boats** and **water** identify the correct sense as **riverbank**.

Benefits:

1. Improves translation accuracy.
2. Preserves semantic relationships.
3. Reduces ambiguity.
4. Produces coherent sentences.
5. Enhances NLP applications such as question answering, information retrieval, and text summarization.

Constraint Analysis Summary (Rubric Alignment)

Criterion	Satisfaction
Problem Understanding	All constraints identified
Constraint Analysis	Gender, number, recency, semantics, coherence applied
Solution Development	Correct reference resolution and responses generated
Application of Concepts	Coreference, dialog acts, WSD, predicate logic used
Justification	Logical explanations provided throughout